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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of Computer Science & Engineering

"A Place to Learn, A Chance to Grow"

Session: 2025-26



VISION

To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.

MISSION

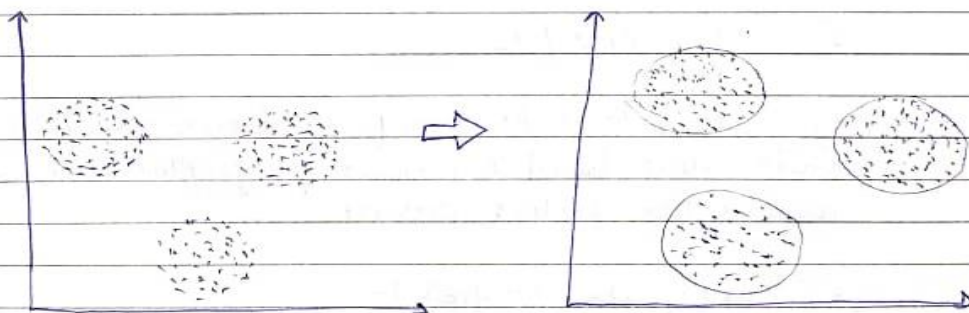
1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities.
2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program.
3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

UNIT – 02

* Clustering :-

- clustering is basically a type of unsupervised learning method.
- Clustering is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group and dissimilar to the data points in other groups.
- It is basically a collection of objects on the basis of similarity and dissimilarity between them.
- For Example.

The data points in the graph below clustered together can be classified into one single group. we can distinguish the clusters, and we can identify that there are 3 clusters in the below figure.



* Applications of clustering in different fields

① Marketing :-

It can be used to characterize & discover clusters into customer segments for marketing purposes.

② Biology :-

It can be used for classification among different species of plants and animals.

③ Libraries :-

It is used in clustering different books on the basis of topics and information.

④ Insurance :-

It is used to acknowledge the customers, their policies and identifying the frauds.

⑤ City planning :-

It is used to make groups of houses and to study their values based on their geographical locations and other factors present.

⑥ Earthquake studies :-

By learning the earthquake-affected areas we can determine the dangerous zones.

Hierarchical Clustering

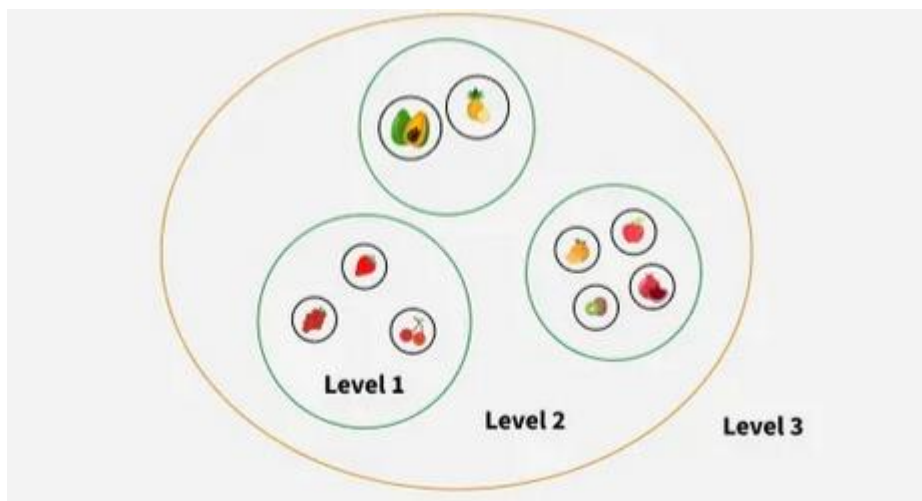
- Hierarchical clustering is an unsupervised learning technique used to group similar data points together based on their Similarity Creating a hierarchy or tree-like structure.
- The Key idea is to begin with each data point as its own Separate Cluster and then Progressively merge or Split them based on their Similarity

- **Example**

Imagine we have four fruits with different weights: an apple (100g), a banana (120g), a cherry (50g) and a grape (30g). Hierarchical clustering starts by treating each fruit as its own group.

- Start with each fruit as its own cluster.
- Merge the closest items: grape (30g) and cherry (50g) are grouped first.
- Next, apple (100g) and banana (120g) are grouped.
- Finally, these two clusters merge into one.

Finally, all the fruits are merged into one large group, showing how hierarchical clustering progressively combines the most similar data points.



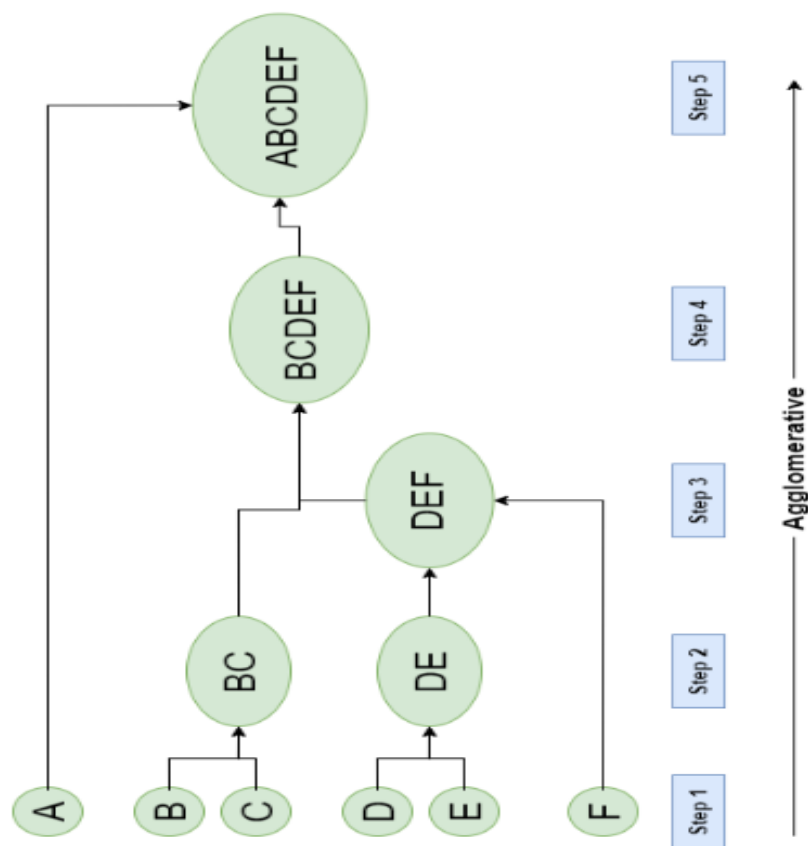
Hierarchical Agglomerative Clustering

- It is also known as the **bottom-up approach** or **hierarchical agglomerative clustering (HAC)**.
- Unlike flat clustering hierarchical clustering provides a structured way to group data.
- This clustering algorithm does not require us to prespecify the number of clusters.
- Bottom-up algorithms treat each data as a singleton cluster at the outset and then successively agglomerate pairs of clusters until all clusters have been merged into a single cluster that contains all data.

Workflow for Hierarchical Agglomerative clustering

1. **Start with individual points:** Each data point is its own cluster. For example, if you have 5 data points you start with 5 clusters each containing just one data point.
2. **Calculate distances between clusters:** Calculate the distance between every pair of clusters. Initially since each cluster has one point this is the distance between the two data points.

3. **Merge the closest clusters:** Identify the two clusters with the smallest distance and merge them into a single cluster.
4. **Update distance matrix:** After merging you now have one less cluster. Recalculate the distances between the new cluster and the remaining clusters.
5. **Repeat steps 3 and 4:** Keep merging the closest clusters and updating the distance matrix until you have only one cluster left.
6. **Create a dendrogram:** As the process continues you can visualize the merging of clusters using a tree-like diagram called a **dendrogram**. It shows the hierarchy of how clusters are merged.



Hierarchical Agglomerative Clustering

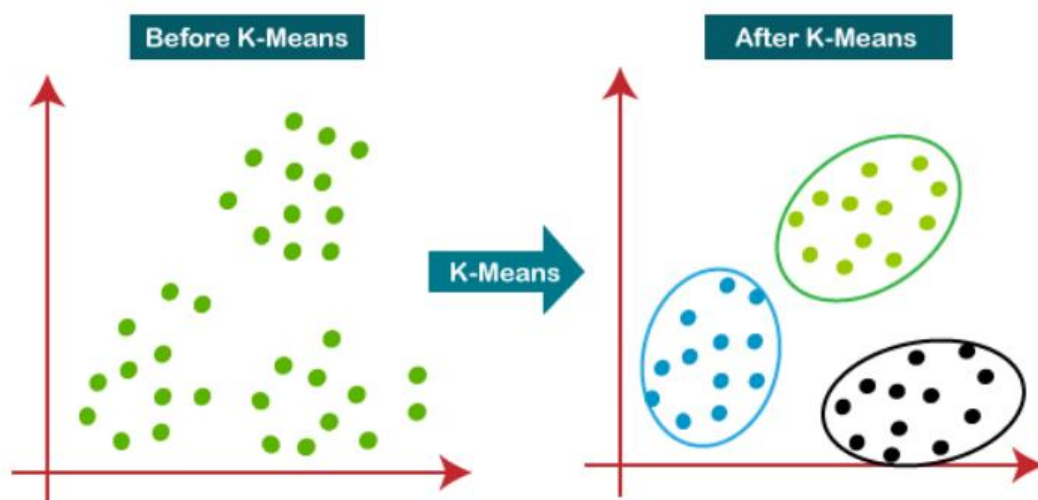
K-Means Clustering:

- K-Means Clustering is an Unsupervised Learning algorithm, which groups the unlabeled dataset into different clusters.
- Here K defines the number of pre-defined clusters that need to be created in the process, as if $K=2$, there will be two clusters, and for $K=3$, there will be three clusters, and so on.
- It is an iterative algorithm that divides the unlabeled dataset into k different clusters in such a way that each dataset belongs only one group that has similar properties.
- It is a centroid-based algorithm, where each cluster is associated with a centroid.

- The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters.
- The algorithm takes the unlabeled dataset as input, divides the dataset into k-number of clusters, and repeats the process until it does not find the best clusters.
- The value of k should be predetermined in this algorithm.
- The k-means clustering algorithm mainly performs two tasks:
 - o Determines the best value for K center points or centroids by an iterative process.
 - o Assigns each data point to its closest k-center. Those data points which are near to the particular k-center, create a cluster.

Hence each cluster has datapoints with some commonalities, and it is away from other clusters.

The below diagram explains the working of the K-means Clustering Algorithm:



How does the K-Means Algorithm Work?

The working of the K-Means algorithm is explained in the below steps:

Step-1: Select the number K to decide the number of clusters.

Step-2: Select random K points or centroids. (It can be other from the input dataset).

Step-3: Assign each data point to their closest centroid, which will form the predefined K clusters.

Step-4: Calculate the variance and place a new centroid of each cluster.

Step-5: Repeat the third steps, which means reassign each datapoint to the new closest centroid of each cluster.

Step-6: If any reassignment occurs, then go to step-4 else go to FINISH.

Step-7: The model is ready.

Batchelor-Wilkins algorithm

- The Batchelor-Wilkins algorithm is a method used in cluster analysis, particularly for determining the optimal number of clusters (K-value) in a dataset, which is often a challenge in algorithms like K-means.
- It helps to solve the Problem of Selecting the best number of natural groupings in the Data.

The step-by-step process of the Batchelor-Wilkins algorithm is as follows:

Initial Clustering:

- Apply a clustering algorithm (e.g., K-means) multiple times with different values of K (for example, 2, 3, 4, 5...).
- This step groups the data points into clusters based on their similarities.

Evaluating Cluster Validity:

- For each K value, measure how good the clustering is using **cluster validity indices**, such as:
 - **Silhouette Score:** Measures how well-separated clusters are.
 - **Davies-Bouldin Index:** Measures cluster similarity.
 - **Calinski-Harabasz Index:** Measures variance between clusters.
- A good clustering means:
 - Points inside the same cluster are close together (compactness).
 - Different clusters are far apart (good separation).

Choose the Best K

- Compare the evaluation scores for all K values.
- Select the K that gives the **best score** (e.g., highest Silhouette Score).
- This represents the **optimal number of clusters**.

Identify Cluster Centers

- With the chosen K, determine the **centroids (central points)** of each cluster.
- These represent the main “center” or “mean point” of the cluster.

Final Clustering

- Assign every data point to the **nearest centroid** based on the chosen K.
- This results in the final, most accurate clustering of the dataset.

BATCHELOR-WILKINS ALGORITHM

